

SAMPLING-UNIT AND COMPOSITION SENSITIVITY OF AN AUDIO DEEPPFAKE LEADERBOARD

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ABSTRACT

ASVspoof 2021 cites Bengio–Mariéthoz for its deepfake significance matrix without disclosing which test variant was used. Reconstructions separate five of six organizer-baseline pairs. With identical EER threshold refitting and all-pair multiplicity, trial-i.i.d. perturbation also gives 5/6, against 0/6 under speaker–attack product weights; a coherent marginal-sum covariance gives 0/6. A pre-access registered, single-source SpoofCeleb check yields 6/6 against 3/6, isolating sampling-unit from source-composition movement. An external fixed-family ASVspoof 5 check meets both registered family criteria (median width inflation $27.27\times$). Separately, 12 predeclared, score-blind composition policies (axes outcome-informed) reverse 6/12 within-generation orderings but 0/16 cross-generation ones. A post-failure search verified by three EER backends finds witnesses for all 12 within-generation pairs, five at total variation $\leq .10$ and the smallest at .010218. A frozen low-EER stress cell has nominal-95% coverage .879/.880/.883, refuting universal calibration; 85.77% of spoof trials lie in blocks with only four or six speakers. We identify fixed-data evaluation-design sensitivity, not corrected population inference: the coarse generation gap is stable over registered policies, while within-generation rankings are not.

Index Terms— Anti-spoofing, audio deepfake detection, evaluation methodology, clustered bootstrap, ranking uncertainty

1. INTRODUCTION

In 2021 the ASVspoof organizers tested every pair of submitted systems for significance, applied a Holm–Bonferroni correction, and published the deepfake task’s systems in four groups, noting that the top two were not separated [1]. We found no uncertainty analysis in three subsequent benchmark reports: the journal overview [2], ASVspoof 5 [3], and the recent multi-benchmark leaderboard [4] contain none of *significance test*, *confidence interval*, *bootstrap*, *p-value* or *standard error*. The sub-point margins on that modern leaderboard are therefore reported without an uncertainty

test; a 2026 benchmark of self-supervised models likewise ranks point EERs without an interval, error bar or test [5]. The 2026 edition adds a calibration study instead; calibration establishes whether a score means what it says, not whether two systems differ.

That matters more than it would if the analyses the field did run had represented the dependence. ASVspoof 2017 reported 95% EER confidence intervals and read their overlap [6], and 2021 tested every pair [1]; both use [7], and the practice then lapsed. Both variants of the cited test [7] treat trials within a class as independent Bernoulli draws; the dependent form accounts for two systems sharing trials, not for trials sharing speakers and attacks. ASVspoof 2021 DF contains 533,928 evaluation trials but only 93 speakers and 110 attacks, and per-attack EERs for SSL-AASIST span 0.4–10.7%. Speaker ICC on bona-fide scores is 0.15–0.77, so repeated trials plainly do not supply independent information. ICC alone does not establish paired error correlation or an EER design effect; those are measured directly below by matched perturbations of the joint scores and then checked externally at family level on ASVspoof 5.

We vary two declared evaluation-design choices: the sampling unit used to perturb paired scores, and the observed source/task composition used to weight them. No statistical method is new—clustered resampling is standard [8, 9, 10]—and the contribution is the measured sensitivity and its audited boundary. The 5/6-to-0/6 change survives matched threshold re-estimation and coherent covariance, while a pre-access registered single-source SpoofCeleb check gives 6/6-to-3/6; explicit composition witnesses reverse within-generation point orderings. A tail stress test refutes low-EER calibration, while metadata reject global speaker–attack exchangeability. The defensible result concerns outputs of specified procedures on released scores, not a replacement population verdict.

Scope. The primary pool comprises eight author or organizer releases with high provenance, not the universe of public re-scores. We analyze all eleven Speech DF Arena re-scores [4] separately. No full-21DF “resolved” label transfers to new speakers or attacks: four source/task blocks have only 4–6

speakers each. Bands and composition radii are descriptive procedure outputs on fixed scores, not population confidence or safety claims.

2. RELATED WORK

This benchmark’s own reporting. ASVspoof 2021 pairs systems with the test of Bengio and Mariéthoz [7] under Holm correction and publishes group structure [1]; Wang and Yamagishi [11] showed that changing only the training seed moves ADD rankings—a variance component our intervals do *not* include.

Clustered resampling. Poh, Martin and Bengio [8] showed that biometric uncertainty is governed by users rather than sample count; Wu et al. carried user-aware resampling into speech [12] and developed a paired correlation-corrected two-system test [9]. Bolle et al. proposed a subsets bootstrap [13], while Fogliato et al. found undercoverage for dyadic 1:1 matching [14]. Our trials instead have one speaker and, for spoof, one attack: two crossed factors [15], a delicate design [16, 17] distinct from nested metrics [18]. Speech bootstrap intervals also include utterance- and block-based forms [19, 10]. Our contribution is the audited fixed-data result: after EER threshold refitting and all-pair multiplicity are aligned, outputs change with the declared perturbation unit. We claim neither cell-level agreement with the published greyscale figure nor population confidence.

Benchmarks that cannot resolve what is claimed on them. Zhuang et al. [20] compute a paired-binomial detectable effect for quantization benchmarks, treating items as independent—the assumption under test here. In audio deepfakes, bona-fide cross testing shows aggregation weights change conclusions [21]; other audits question representations [22] and corpus overlap [23]. None gives pairwise composition-reversal witnesses or names a dependence unit for uncertainty. Simultaneous rank intervals are established [24]; adjacency is score-selected, so we control multiplicity over the whole family.

3. METHOD

3.1. Identified object: fixed-data procedure sensitivity

Two deterministic systems on a frozen trial list have *no* sampling uncertainty; an interval exists only after declaring what was sampled. The organizers’ test treats trials within class as independent draws [7]. Our original working target instead treated speakers and attacks as exchangeable with a represented evaluation design. Metadata falsify a global version of that design: source/task stratification leaves almost no empty cells, yet the VCC blocks contain only 4/4/4/6 spoof speakers. We therefore do not replace the organizers’ population claim with another unidentified one. The identified object is

how paired Δ EER outputs on the fixed score set change under declared trial-i.i.d. and speaker–attack perturbation laws; confidence language is reserved for a future target with a defensible sampling law and replication.

3.2. Clustered bootstrap and simulation-checked variants

Speakers and attacks are resampled independently with replacement and trial weights multiply the two counts; bona fide trials carry no attack label and so receive speaker weights only. The primary output is the two-way percentile bootstrap. Exact delete-one refits separately compute $\Sigma_s + \Sigma_a - \Sigma_{sa}$ over 93 speakers, 110 spoof attacks and 1,062 observed spoof speaker $\times$ attack cells [15]. Its former pairwise floor $\max(V_s + V_a - V_{sa}, V_s, V_a)$ did not define a joint covariance and is retired from simultaneous labels. The replacement corroborator uses the single PSD matrix $\Sigma_+ = \Sigma_s + \Sigma_a$: it dominates both marginals and the raw exact-cell matrix in Loewner order, and the same Σ_+ supplies every pair SE and the Gaussian max- t critical value ($q = 2.878$). It deliberately retains the cell overlap and is a coherent marginal-sum sensitivity covariance, not an exact multiway estimator or coverage claim. Product and coherent outputs agree on all 28 zero-exclusion labels (18/28; organizer 0/6). A smooth influence-function CRVE failed its prospectively frozen internal finite-refit gates on every pair and is excluded.

3.3. All-pair multiplicity for procedure bands

From $B=5000$ clustered replicates we form sup- t simultaneous procedure bands [25]: the critical value is the 95th percentile of $\max_p |\Delta_p^{(b)} - \bar{\Delta}_p|/s_p$ over replicates, applied to every pair. Rank sets are derived mechanically from those bands. Because the required sampling law is not identified for full 21DF, neither is called a population confidence set. A bootstrap rank percentile is separately released as a stability interval.

Reconstructing the cited test. The overview does not disclose which Bengio–Mariéthoz variant produced its greyscale matrix, so we make no cell-level attribution. With each system’s non-interpolated EER threshold selected on the evaluation set, the independent and paired-decision reconstructions separate the same five of six baseline pairs. Holm is step-down, but the result is family-size independent: in the independent reconstruction the weakest significant pair has $p = 9.7 \times 10^{-6} < 0.05/561$, while the sixth has $p = 0.217 > 0.05$. Because the cited derivation assumes thresholds fixed outside the test set, these are reconstructions of its adaptation to EER rather than exact finite-sample tests.

3.4. Coverage check

A bivariate random-effects model generates scores *on the real trial index*—the actual speaker and attack of every trial,

preserving the sparse unbalanced incidence but imposing the exchangeability law that the real source/task structure later rejects. The prospectively frozen internal 48-cell grid crosses organizer-like/low EER, interaction shares 0–.75, Gaussian/Student- $t(5)$ effects and $\Delta \in \{0, .5, 2\}$, with high-precision population truth. The three then-candidate variants cover within $[\text{.92}, \text{.98}]$ in all 24 organizer-like model cells (ranges .928–.957 exact-cell and .946–.968 product). In low EER their minima are .845/.845/.855, and all three fall below .90 together in 11 of 24 cells. The frozen reading rule is therefore Refuted. An oracle constant-SD diagnostic stays at or above .944, while replicate-specific SE shrinks in extreme negative-tail samples; the defect is asymmetric non-pivotality, not average variance scale. Passing organizer-like cells validates implementation behavior under that DGP, not the DGP as a model of 21DF.

3.5. Composition sensitivity on fixed scores

Three bona-fide policies (empirical, source-balanced, source-speaker-balanced) and four spoof policies (empirical through attack-family-cell balanced) form a frozen 12-policy factorial. All trial weights remain positive and class-normalized. For each pair we report the policy range and registered empirical-to-policy paths; r_{TV} is the larger class-conditional total-variation shift. These are deterministic estimands, not confidence intervals. After two failed numerical implementations, a separately frozen post-failure search may only exhibit constructive upper bounds: a witness must reverse under count-canonical, direct-trial and tie-safe EER evaluators. Search failure is not stability; a reported bound is not a global minimum.

4. EXPERIMENTS

Setup. The primary pool contains eight high-provenance ASVspoof 2021 DF releases: four self-supervised systems (XLSR-Mamba [26], XLS-R+SLS [27], XLSR-Conformer [28], SSL-AASIST [29]) from their authors and four organizer baselines (RawNet2, LFCC-LCNN, LFCC-GMM, CQCC-GMM) from the challenge key package. We use the 533,928 eval-phase trials of 611,829 released. Pooled EERs run 1.88–2.85 for the self-supervised systems and 22.38–25.56 for the baselines, reproducing published values where they exist. Pairwise intervals use $B=5000$ clustered replicates (seed 20260817); the 48-cell coverage stress test uses $B=500$ per replicate over $R=1000$ replicates per cell (seed 20260824).

Sampling-unit sensitivity (Fig. 1). Both published-test reconstructions separate the same five of six organizer pairs; we claim no cell-level agreement with the greyscale figure. A matched all-28-pair max- t bootstrap refits the same non-interpolated weighted EER in every replicate: trial-i.i.d. perturbation gives 26/28 and 5/6 organizer exclusions, against

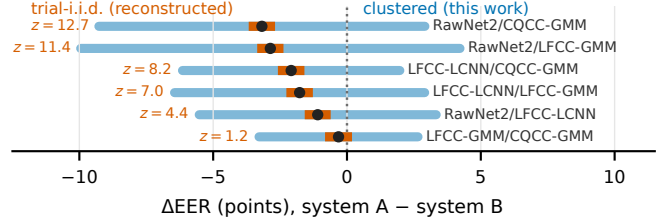


Fig. 1. The six baseline pairs, which appear in both the published significance figure and our pool. Narrow bars: the independent Bengio–Mariéthoz reconstruction (its z at left); paired-decision reconstruction gives the same five verdicts. Wide bars: the simultaneous speaker–attack procedure output. Dots: point estimates. All six straddle zero after reweighting.

18/28 and 0/6 under speaker–attack product weights. Coherent marginal-sum covariance and a transparently post-failure composition-fixed control also give 18/28 and 0/6; the latter conditions only zero-support stratum draws and exactly restores each class-by-source/task mass. Thus neither threshold treatment nor that composition shift explains the contrast. The widest 3.18-point gap has product band $[-9.27, 2.91]$ and coherent band $[-8.76, 2.40]$.

On the complete 91,130-trial single-source SpoofCeleb evaluation grid [30], four fixed off-domain detectors give 6/6 trial-i.i.d. versus 3/6 product/source exclusions under a rule registered before access. Every base product has one bona-fide and nine spoof renditions under one source label, so composition is fixed by construction; the three changed decisions isolate sampling-unit sensitivity from source-mixture movement. Gate 2 later found that the original seal omitted the actual scorer endpoint. A disclosed post-result rerun from clean, hash-bound commits reproduced three score files byte-for-byte; XLSR-Mamba moved by at most 1.43×10^{-6} , with all EERs, decisions and the registered endpoint unchanged.

What the benchmark supports (Table 1). All six organizer pairs straddle zero under product and coherent marginal-sum perturbations despite a 3.18-point span. RawNet2–CQCC-GMM, LFCC-LCNN–LFCC-GMM and LFCC-LCNN–CQCC-GMM become zero-excluding in leave-one-corpus-out refits, so the organizer result is not composition-robust. The 16 cross-generation gaps are 19.5–23.7 points and survive every deletion. Two modern zero-exclusions at 0.97 and 0.94 points are withdrawn because low-EER stress is anti-conservative and both disappear after source deletion; adjacent modern margins are only 0.03, 0.36 and 0.58 points.

Composition robustness. Under the 12 registered score-blind policies, 6/12 within-generation orderings reverse and 0/16 cross-generation orderings do; only one registered path reverses at $r_{TV} \leq .10$, yielding the frozen classification *Local composition fragility*. The later constructive search cannot

Table 1. Within-generation pairs: empirical gap and product-perturbation pointwise percentile half-width (EER points; the simultaneous bands are reported in the text), plus the smallest verified constructive r_{TV} upper bound. A bound is a found reversal, not a minimum or confidence limit.

pair		gap	half-w.	$r_{TV} \leq$	
XLSR-Mamba	XLS-R+SLS	0.032	0.452	0.011	
XLSR-Mamba	XLSR-Conf	0.389	0.619	0.344	
XLSR-Mamba	SSL-AAS	0.968	0.555	0.246	
XLS-R+SLS	XLSR-Conf	0.357	0.752	0.172	
XLS-R+SLS	SSL-AAS	0.935	0.561	0.434	
XLSR-Conf	SSL-AAS	0.578	0.674	0.166	
RawNet2		LFCC-LCNN	1.094	2.868	0.069
RawNet2		LFCC-GMM	2.864	4.638	0.094
RawNet2		CQCC-GMM	3.180	3.952	0.112
LFCC-LCNN		LFCC-GMM	1.770	3.074	0.087
LFCC-LCNN		CQCC-GMM	2.086	2.636	0.104
LFCC-GMM		CQCC-GMM	0.316	1.996	0.034

revise that classification: it verifies 729 directions and bounds all 12 within-generation reversals from above, five at $r_{TV} \leq .10$. XLSR-Mamba versus XLS-R+SLS reverses at .010218. All three EER evaluators agree, but the delta at that witness is tiny; the result is local sensitivity, not practical superiority of either system. No cross-generation witness was found, which is a search outcome rather than proof of global stability.

Independent re-score layer. All eleven complete Speech DF Arena files map one-to-one to the same 533,928 trials, but remain a separate stratum because they are re-scores and same-named implementations need not match: Arena RawNet2 has 40.67% EER against 22.38% in the organizer release and score correlation 0.36. Applying the same procedure descriptively to the full 55-pair family changes the count of zero-excluding bands from 52 under trial-i.i.d. inference to 38 after speaker $\times$ attack resampling; two score-adjacent edges change labels, and the median clustered/i.i.d. pointwise-width ratio is 8.45. These are provenance and sensitivity measurements, not coverage-validated Arena confidence claims.

Speakers are not exchangeable either. The three bona-fide sources contain 67/12/14 speakers; corpus-grouped delete-one influence contributes 9–48% of its finite- A sum of squares (median 26%). Against a 2.2% permutation null, all 12 pairs survive Holm ($p_{\text{Holm}} \leq .041$): treating corpus as exchangeable leaves speaker bands anti-conservative. Leave-one-corpus-out exact-cell refits preserve all 16 cross-generation verdicts, remove both within-SSL resolutions and add three baseline ones. Thus block separation survives provenance changes, while every within-generation resolution is conditional on them.

External family-level check. ASVspoof 5 replaces about a hundred studio-recorded speakers with 737 crowdsourced ones and cuts attacks from 110 to 16 [3]. We froze SSL-AASIST, AASIST, XLS-R+SLS and XLSR-Mamba (EER

16.25, 35.53, 18.76 and 14.40%) over all 680,774 trials (367 target and 370 bona-fide-only speakers). Criterion A required at least one trial-i.i.d. exclusion to disappear; B required median width inflation ≥ 2 . Both hold. Role-stratified speaker-attack perturbation changes SSL-AASIST versus XLS-R+SLS from $[-2.657, -2.378]$ to $[-6.274, 1.238]$ EER points. Across the fixed six-pair family, bands are $11.23\text{--}35.90\times$ wider (median $27.27\times$); the focal ratio is $26.92\times$. This is not pair-level replication: on 21DF that pair excludes zero under both arms, and the 5/6-to-0/6 organizer family has no ASV5 analogue. The acquisition-law gate is NO-GO; these are fixed-roster procedure outputs, not population confidence or significance.

Measured width inflation. On the real score files, simultaneous product bands are $4.13\text{--}11.36\times$ wider than matched trial-i.i.d. bands over all 28 pairs (median $8.14\times$); both arms refit the same EER threshold. In a witnessed $R=200$ run under the fitted organizer-like Gaussian DGP on the real incidence, nominal-95% trial-i.i.d. coverage is 18.5% (Wilson 95% interval 13.7–24.5%), versus 97.5–98.0% for the three clustered candidates (combined endpoint range 94.3–99.2%). This calibrates procedures under the imposed DGP, not the DGP as a model of 21DF or its unidentified superpopulation.

5. DISCUSSION

For reporting, and limits. Plans, deviations, hashes and receipts are at github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty. Legacy SSL-AASIST/AASIST NPZs lack historical run provenance, and the historical freeze lacks an independently verifiable timestamp. Report speaker and attack incidence plus composition, not trial counts: bands require a sampling law and coverage; radii require a policy family and witness. We withdraw full-21DF population labels, modern rank-confidence claims and formal Arena bands. Zero-straddling is sensitivity, not equality.

6. CONCLUSION

Matched 21DF perturbations yield 5/6 versus 0/6 organizer exclusions, and registered single-source SpoofCeleb yields 6/6 versus 3/6; composition reverses 6/12 within- but 0/16 cross-generation orderings. These are fixed-score sensitivities, not population inference; that needs new independently sampled units.

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